Computer Science & Information Technology

Computer Networks 2

Framing & Error Control

DPP: 1

Q1 A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the input bit-string is 10111110111, then the output bit-string after stuffing is

(A) 10111110111 (B) 1011111111
(C) 101111100111 (D) 1011111001110

Q2 A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 10111110111, then the input bit-string is

(A) 10111110111 (B) 1011111111
(C) 101111100111 (D) 1011111001110

Q3 Consider 2D parity scheme for error detection, let suppose size of data block matrix (including parity bits) is 5×6. Count total number of data bits in the data block (excluding parity bits) ?

Q4 Consider 2D parity scheme for error detection with even parity is used and size of matrix is 4×5. Let suppose 20 bits block (contains 12 data bits) is 0×A36EF (represented in hexadecimal) received by receiver. The received block contains n corrupted bits. What should be minimum possible value of n ?

Q5 In hamming code, how many minimum number of parity bits required for 12 data bits?

Q6 Consider even parity in hamming code and first parity is placed at least significant bit position, the data bits 1101 is encoded as:

(A) 1101110 (B) 1100100
(C) 1100110 (D) 1101101

Answer Key

Q1	(C)	Q4	1~1
Q2	(B)	Q5	5~5
Q3	20~20	Q6	(C)

Hints & Solutions

Q1 Text Solution:

Frame delimiter = 01111110

Data = 10111110111

Transmitted data = 101111100111

Q2 Text Solution:

Frame Delimiter = 01111110

Transmitted Data = 10111110111

Data = 1011111111

Q3 Text Solution:

2D parity

Block size = 5 × 6

No. of data bits = (5 – 1) × (6 – 1)

= 4 × 5 = 20

Q4 Text Solution:

Received data = O×A36EF (Even parity)

10100011011011101111

$$\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ \hline 0 & 1 & 1 & 1 & 1 \end{array}$$

Min^m one bit Error

Ans = 1

Q5 Text Solution:

No. of data bits(m) = 12

No. of parity bits(r) = ?

$2^r > (m + r)$

Min^m value for r = 5

Q6 Text Solution:

Data = 1101, Even parity

$$\frac{1}{7} \quad \frac{1}{5} \quad \frac{0}{5} \quad \frac{0}{4} \quad \frac{1}{3} \quad \frac{1}{2} \quad \frac{0}{1}$$